

HDOC Day-25

Decreasing Patterns Using Nested Loops and Switch-Case in C

Aim

To write a menu-driven C program using switch-case, nested loops and separate void functions to print the four decreasing patterns given in the problem without using arrays.

Theory / How the Program Works

A **do-while loop** keeps displaying the menu until Exit is selected. A **switch statement** calls the function for the selected pattern. Each pattern is written as a separate function with no return value. The outer loop controls the rows and decreases from n to 1, while the inner loop prints the required stars or numbers. The row value n is validated so that only positive values are accepted.

Menu

Choice	Pattern
1	Decreasing right-angled triangle
2	Decreasing natural-number pattern
3	Decreasing repeated-number pattern
4	Reverse-number triangle
5	Exit

Algorithm

1. Start and declare the menu choice.
2. Display the decreasing-pattern menu and read the user's choice.
3. Use switch-case: choices 1-4 call their respective functions, choice 5 exits, and default displays an invalid-choice message.
4. In each pattern function, input n. If $n \leq 0$, display an error and repeat until n is positive.
5. Pattern 1: for i from n down to 1, print i asterisks.
6. Pattern 2: for i from n down to 1, print natural numbers from 1 to i.
7. Pattern 3: for i from n down to 1, print the value i exactly i times.
8. Pattern 4: for i from n down to 1, print numbers from i down to 1.
9. After completing the selected pattern, return to the main menu.
10. Repeat until the user selects Exit.
11. Stop.

Test-Case Table

Test	Choice / Input	Expected result
1	1, n=4	Prints 4, 3, 2 and 1 asterisks on successive rows.
2	2, n=4	Prints 1 2 3 4 / 1 2 3 / 1 2 / 1.
3	3, n=4	Prints 4 4 4 4 / 3 3 3 / 2 2 / 1.
4	4, n=4	Prints 4 3 2 1 / 3 2 1 / 2 1 / 1.

Test	Choice / Input	Expected result
5	1, n=0 then 4	Rejects 0, then accepts 4 and prints the pattern.
6	choice=8	Displays invalid-choice message and repeats menu.
7	choice=5	Exits the program.

C Program

```
#include <stdio.h>

void starTriangle(void);
void naturalNumberPattern(void);
void repeatedNumberPattern(void);
void reverseNumberTriangle(void);

int main()
{
    int choice;

    do
    {
        printf("\n==== DECREASING PATTERNS MENU =====\n");
        printf("1. Decreasing right-angled triangle\n");
        printf("2. Decreasing natural-number pattern\n");
        printf("3. Decreasing repeated-number pattern\n");
        printf("4. Reverse-number triangle\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                starTriangle();
                break;
            case 2:
                naturalNumberPattern();
                break;
            case 3:
                repeatedNumberPattern();
                break;
            case 4:
                reverseNumberTriangle();
                break;
            case 5:
                printf("Exiting program...\n");
                break;
            default:
                printf("Invalid choice! Enter a value from 1 to 5.\n");
        }

    } while (choice != 5);

    return 0;
}

void starTriangle(void)
{
    int n, i, j;

    do
    {
        printf("Enter number of rows: ");
        scanf("%d", &n);

        if (n <= 0)
            printf("Invalid input! Rows must be positive.\n");

    } while (n <= 0);

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("* ");

        printf("\n");
    }
}

void naturalNumberPattern(void)
{
    int n, i, j;

    do
    {
        printf("Enter number of rows: ");
        scanf("%d", &n);
        if (n <= 0)
```

```

        printf("Invalid input! Rows must be positive.\n");
    } while (n <= 0);

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("%d ", j);

        printf("\n");
    }
}

void repeatedNumberPattern(void)
{
    int n, i, j;

    do
    {
        printf("Enter number of rows: ");
        scanf("%d", &n);

        if (n <= 0)
            printf("Invalid input! Rows must be positive.\n");

    } while (n <= 0);

    for (i = n; i >= 1; i--)
    {
        for (j = 1; j <= i; j++)
            printf("%d ", i);

        printf("\n");
    }
}

void reverseNumberTriangle(void)
{
    int n, i, j;

    do
    {
        printf("Enter number of rows: ");
        scanf("%d", &n);

        if (n <= 0)
            printf("Invalid input! Rows must be positive.\n");

    } while (n <= 0);

    for (i = n; i >= 1; i--)
    {
        for (j = i; j >= 1; j--)
            printf("%d ", j);

        printf("\n");
    }
}

```

Compilation and Execution

The program was compiled using **gcc -Wall -Wextra hdoc_day25.c -o hdoc_day25**. Compilation completed successfully. The program was executed using $n = 4$ for all four patterns and was also tested with an invalid menu choice.

Sample Output

```
===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: Enter number of rows: * * * *
* * *
* *
*

===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: Enter number of rows: 1 2 3 4
1 2 3
1 2
1

===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: Enter number of rows: 4 4 4 4
3 3 3
2 2
1

===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: Enter number of rows: 4 3 2 1
3 2 1
2 1
1

===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: Invalid choice! Enter a value from 1 to 5.

===== DECREASING PATTERNS MENU =====
1. Decreasing right-angled triangle
2. Decreasing natural-number pattern
3. Decreasing repeated-number pattern
4. Reverse-number triangle
5. Exit
Enter your choice: Exiting program...
```

Result

The menu-driven C program was successfully written, compiled and executed. All four decreasing patterns are generated correctly using nested loops and separate void functions. Invalid row values and invalid menu choices are handled, and the menu continues until Exit is selected.